Equivalent expressions — Part 2 answer key

#	ID	Answer	Difficulty
2.1	c0558845-d200-4e15-8b86-f37c4c3c5484	D	Medium
2.2	2aeea10a-c9ed-426c-912a-939dedc77aac	B	Medium
2.3	891c7c48-227c-4992-acdb-e5e19b360217	A	Medium
2.4	022343-DC	C	Medium
2.5	029490-DC	B	Medium
2.6	74b2dded-7f3f-4e1c-ba2e-dada922f008c	B	Medium
2.7	d228702c-741e-49d7-ad2d-527046f58b94	B	Medium
2.8	00664-DC	B	Medium
2.9	08064-DC	C	Medium
2.10	07749-DC	B	Medium
2.11	070623-DC	?	Medium
2.12	08126-DC	B	Medium
2.13	94498a74-a763-40c5-b248-608f7ff7f6e4	A	Medium
2.14	016475-DC	B	Medium
2.15	74a8d55a-4db1-41ce-9d72-b5498e6a1d89	6	Medium
2.16	022234-DC	B	Medium
2.17	05487-DC	A	Medium
2.18	029468-DC	D	Medium
2.19	05799-DC	B	Medium
2.20	05410-DC	C	Medium

2.21	09342-DC	B	Medium
2.22	09133-DC	?	Medium
2.23	3c39fa3c-f27f-4029-8473-2bb4b3ca5672	B	Medium
2.24	82103492-4feb-429c-a10f-a71972d8a35d	D	Medium
2.25	00538-DC	D	Hard
2.26	025860-DC	C	Hard
2.27	09413-DC	D	Hard
2.28	1775076a-9ccb-4b4b-982d-4765d16527dc	27	Hard
2.29	070622-DC	?	Hard
2.30	44b4a3f1-1823-4c9d-984b-a4ea335c95ad	8	Hard
2.31	05695-DC	D	Hard
2.32	aed0b60c-1ac2-47c0-8e11-047b2c3d1b5e	.0177, .0178, 4/225	Hard
2.33	11189-DC	C	Hard
2.34	08605-DC	?	Hard
2.35	f85f6f30-eece-4b2b-bbc1-5c3690c1998f	D	Hard
2.36	8c278545-5668-4346-b11a-ea571647594f	C	Hard
2.37	01383d90-cb64-4777-9f89-45b8dfe8c286	.09, 9/100	Hard
2.38	b76e28cc-fee7-47dc-ae66-119a52a95d28	D	Hard
2.39	7c3cceff-5961-4825-a23b-44ce88420238	C	Hard
2.40	022991-DC	C	Hard
2.41	924baff1-d1d9-4091-910f-6e75684ca3a2	15.5, 31/2	Hard
2.42	16d897b1-c597-4b7a-b4ee-ecba3515085e	D	Hard
2.43	763bb625-3667-4ea6-af9f-195d80193bdb	.5061, .5062, 41/81	Hard

2.44	00489-DC	D	Hard
2.45	00094-DC	D	Hard
2.46	e7cb9147-ae39-47e9-a4cb-da0cfbf0b250	361/8, 45.12, 45.13	Hard
2.47	77bede4f-84ec-4063-9b73-a0d1bd0cff6b	B	Hard
2.48	262f0667-b95d-4adf-bdd9-ff689f8a3811	.2916, .2917, 7/24	Hard
2.49	902558e4-018d-442f-a571-ef4e6c7c9fae	A	Hard
2.50	08243-DC	B	Hard
2.51	028934-DC	A	Hard
2.52	4374fa03-6222-4e03-a842-f2d7a77397bb	D	Hard
2.53	6291292e-680f-4ffb-9f2e-4bf638eaee8d	D	Hard
2.54	08224-DC	C	Hard
2.55	baae72cd-ec86-42e7-81a6-5f31c096baea	-419	Hard
2.56	4feb8c4d-eb86-4f9e-a21a-b22033b4381b	A	Hard
2.57	08128-DC	C	Hard
2.58	4b8535d8-7c5d-4bfd-9256-245ea6bb505f	D	Hard

Rationales

▼ 2.1 — D

Choice D is correct. Since $8x$ is a common factor of each term in the given expression, the expression can be rewritten as $8xx^9 - x^8 + 11$.

Choice A is incorrect. This expression is equivalent to $7x^{11} - 7x^{10} + 87x^2$.

Choice B is incorrect. This expression is equivalent to $8^{10}x - 8^9x + 88x$.

Choice C is incorrect. This expression is equivalent to $8x^{11} - 8x^{10} + 88x^2$.

▼ 2.2 — B

Choice B is correct. It's given that $f(x) = 2x + 3$, $g(x) = 7x - 2$, and $h(x) = 5x + 6$. It's also given that $fx \cdot gx - hx = ax^2 + bx + c$, where a, b, and c are constants. Substituting $2x + 3$ for $f(x)$, $7x - 2$ for $g(x)$, and $5x + 6$ for $h(x)$ in this equation yields $(2x + 3)(7x - 2) - (5x + 6) = ax^2 + bx + c$. Applying the distributive property to the left-hand side of this equation yields $(2x)(7x) + (2x)(-2) + (3)(7x) + (3)(-2) + (-1)(5x) + (-1)(6) = ax^2 + bx + c$, or $14x^2 - 4x + 21x - 6 - 5x - 6 = ax^2 + bx + c$. Combining like terms on the left-hand side of this equation yields $14x^2 + 12x - 12 = ax^2 + bx + c$. It follows that the value of b is 12.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.3 — A

Choice A is correct. Applying the distributive property, the given expression can be written as $7x^3 + 7x - 6x^3 + 3x$. Grouping like terms in this expression yields $7x^3 - 6x^3 + 7x + 3x$. Combining like terms in this expression yields $x^3 + 10x$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.4 — C

Choice C is correct. Distributing the negative sign to the terms in the second parentheses yields $(2x+3)-x+7$. This expression can be rewritten as $2x-x+3+7$. Combining like terms results in $x+10$.

Choice A is incorrect and may result from not distributing the negative sign to the 7. Choice B is incorrect and may result from adding $(x-7)$ to $2x+3$ instead of subtracting $(x-7)$. Choice D is incorrect and may result from adding the product of $2x$ and x to the product of 3 and 7.

▼ 2.5 — B

Choice B is correct. Using the distributive property, the given expression can be rewritten as $4x^3-5x^2+3-6x^3-2x^2+x$. Combining like terms, this expression can be rewritten as $(4-6)x^3+(-5-2)x^2+x+3$, which is equivalent to $-2x^3-7x^2+x+3$.

Choices A, C, and D are incorrect and may result from an error when applying the distributive property or an error when combining like terms.

▼ 2.6 — B

Choice B is correct. The given expression is equivalent to $8x^3 + 8 - x^3 - -2$, or $8x^3 + 8 - x^3 + 2$. Combining like terms in this expression yields $7x^3 + 10$.

Choice A is incorrect. This expression is equivalent to $8x^3 + 8 - 2$, not $8x^3 + 8 - x^3 - 2$.

Choice C is incorrect. This expression is equivalent to $8x^3 + 8 - -2$, not $8x^3 + 8 - x^3 - 2$.

Choice D is incorrect. This expression is equivalent to $8x^3 + 8 - x^3 + 2$, not $8x^3 + 8 - x^3 - 2$.

▼ 2.7 — B

Choice B is correct. The given expression has a common factor of 2 in the denominator, so the expression can be rewritten as $\frac{8xx - 7 - 3x - 7}{2x - 7}$. The three terms in this expression have a common factor of $x - 7$. Since it's given that $x > 7$, x can't be equal to 7, which means $x - 7$ can't be equal to 0. Therefore, each term in the expression, $\frac{8xx - 7 - 3x - 7}{2x - 7}$, can be divided by $x - 7$, which gives $\frac{8x - 3}{2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.8 — B

Choice B is correct. The given quadratic expression is in standard form, and each answer choice is in vertex form. Completing the square converts the expression from standard form to vertex form. The first step is to rewrite the expression as follows: $x^2+6x+4=x^2+6x+9+4-9$. The first three terms of the revised expression can be

rewritten as a perfect square as follows: $x^2+6x+9+4-9=(x+3)^2+4-9$. Combining the constant terms gives $(x+3)^2-5$.

Choice A is incorrect. Squaring the binomial and simplifying the expression in choice A gives $x^2+6x+9+5$. Combining like terms gives $x^2+6x+14$, not x^2+6x+4. Choice C is incorrect. Squaring the binomial and simplifying the expression in choice C gives $x^2-6x+9+5$. Combining like terms gives $x^2-6x+14$, not x^2+6x+4. Choice D is incorrect. Squaring the binomial and simplifying the expression in choice D gives $x^2-6x+9-5$. Combining like terms gives x^2-6x+4, not x^2+6x+4.

▼ 2.9 — C

Choice C is correct. The expression can be written as a difference of squares $x^2 - y^2$, which can be factored as (x + y)(x – y). Here, $y^2 = 5$, so $y=\sqrt{5}$, and the expression therefore factors as $(x+\sqrt{5})(x-\sqrt{5})$.

Choices A and B are incorrect and may result from misunderstanding how to factor a difference of squares. Choice D is incorrect; (x + 5)(x – 1) can be rewritten as $x^2 + 4x – 5$, which is not equivalent to the original expression.

▼ 2.10 — B

Choice B is correct. It's given that $x^2=a+b$ and $y^2=a+c$. Using the distributive property, the expression $(x^2-y^2)^2$ can be rewritten as $(x^2)^2-2x^2y^2+(y^2)^2$. Substituting $a+b$ and $a+c$ for x^2 and y^2, respectively, in this expression yields $(a+b)^2-2((a+b)(a+c))+(a+c)^2$. Expanding this expression yields $(a^2+2ab+b^2)-(2a^2+2bc+2ac+2ab)+(a^2+2ac+c^2)$. Combining like terms, this expression can be rewritten as $b^2-2bc+c^2$.

Choices A, C, and D are incorrect and may result from an error in using the distributive property, substituting, or combining like terms.

▼ 2.11 — ?

The correct answer is $\frac{5}{2}$. The expression $\left(\frac{1}{2}x+\frac{3}{2}\right)\left(\frac{3}{2}x+\frac{1}{2}\right)$ can be written in the form ax^2+bx+c, where a, b, and c are constants, by multiplying out the expression using the distributive property of multiplication over addition. The result is $\left(\frac{1}{2}x\right)\left(\frac{3}{2}x\right)+\left(\frac{1}{2}x\right)\left(\frac{1}{2}\right)+\left(\frac{3}{2}\right)\left(\frac{3}{2}x\right)+\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)$. This expression can be rewritten by multiplying as indicated to give $\frac{3}{4}x^2+\frac{1}{4}x+\frac{9}{4}x+\frac{3}{4}$, which can be simplified to $\frac{3}{4}x^2+\frac{10}{4}x+\frac{3}{4}$, or $\frac{3}{4}x^2+\frac{5}{2}x+\frac{3}{4}$. This is in the form ax^2+bx+c, where the value of b is $\frac{5}{2}$. Note that 5/2 and 2.5 are examples of ways to enter a correct answer.

▼ 2.12 — B

Choice B is correct. One of the properties of radicals is $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$. Thus, the given expression can be rewritten as $\sqrt[3]{x^3} \cdot \sqrt[3]{y^6}$. Simplifying by taking the cube root of each part gives $x^1 \cdot y^2$, or xy^2.

Choices A, C, and D are incorrect and may be the result of incorrect application of the properties of exponents and radicals.

▼ 2.13 — A

Choice A is correct. The given expression $\frac{4x+1}{x-9}$ can be rewritten as $\frac{4x-36+36+1}{x-9}$, which is equivalent to $\frac{4x-36}{x-9}+\frac{37}{x-9}$. Factoring 4 from the numerator of the first fraction yields $\frac{4(x-9)}{x-9}+\frac{37}{x-9}$. Since $x > 9$, this expression is equivalent to $4+\frac{37}{x-9}$. Therefore, $\frac{4x+1}{x-9}$ is equivalent to $4+\frac{37}{x-9}$.

Choice B is incorrect. This expression is equivalent to $\frac{4x-1}{x-9}$, not $\frac{4x+1}{x-9}$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.14 — B

Choice B is correct. The term x^4 can be factored as $(x^2)(x^2)$. Factoring –6 as $(2)(-3)$ yields values that add to –1, the coefficient of x^2 in the expression.

Choices A, C, and D are incorrect and may result from finding factors of –6 that don't add to the coefficient of x^2 in the original expression.

▼ 2.15 — 6

The correct answer is 6. Applying the distributive property to the expression $ry^4 15y - 9$ yields $15ry^5 - 9ry^4$. Since $90y^5 - 54y^4$ is equivalent to $ry^4 15y - 9$, it follows that $90y^5 - 54y^4$ is also equivalent to $15ry^5 - 9ry^4$. Since these expressions are equivalent, it follows that corresponding coefficients are equivalent. Therefore, $90 = 15r$ and $-54 = -9r$. Solving either of these equations for r will yield the value of r. Dividing both sides of $90 = 15r$ by 15 yields $6 = r$. Therefore, the value of r is 6.

▼ 2.16 — B

Choice B is correct. It's given that $p = 3x + 4$ and $v = x + 5$. Substituting the values for p and v into the expression $pv - 2p + v$ yields $(3x + 4)(x + 5) - 2(3x + 4) + x + 5$. Multiplying the terms $(3x + 4)(x + 5)$ yields $3x^2 + 4x + 15x + 20$. Using the distributive property to rewrite $-2(3x + 4)$ yields $-6x - 8$. Therefore, the entire expression can be represented as $3x^2 + 4x + 15x + 20 - 6x - 8 + x + 5$. Combining like terms yields $3x^2 + 14x + 17$.

Choice A is incorrect and may result from subtracting, instead of adding, the term $x + 5$. Choice C is incorrect. This is the result of multiplying the terms $(3x + 4)(x + 5)$. Choice D is incorrect and may result from distributing 2, instead of -2, to the term $3x + 4$.

▼ 2.17 — A

Choice A is correct. By distributing the minus sign through the expression $\left(\frac{2}{3}x - 5\right)$, the given expression can be rewritten as $\left(\frac{1}{2}x + 3\right) - \frac{2}{3}x + 5$, which is equivalent to $\frac{1}{2}x - \frac{2}{3}x + 3 + 5$. Combining like terms gives $\left(\frac{1}{2} - \frac{2}{3}\right)x + (3 + 5)$, or $-\frac{1}{6}x + 8$.

Choice B is incorrect and may be the result of failing to distribute the minus sign appropriately through the second term and simplifying the expression $\frac{1}{2}x+3-\frac{2}{3}x-5$. Choice C is incorrect and may be the result of multiplying the expressions $\left(\frac{1}{2}x+3\right)$ and $\left(-\frac{2}{3}x+5\right)$. Choice D is incorrect and may be the result of multiplying the expressions $\left(\frac{1}{2}x+3\right)$ and $\left(-\frac{2}{3}x-5\right)$.

▼ 2.18 — D

Choice D is correct. Adding the two expressions yields $3x^4+2x^3+4x^4+7x^3$. Because the pair of terms $3x^4$ and $4x^4$ and the pair of terms $2x^3$ and $7x^3$ each contain the same variable raised to the same power, they are like terms and can be combined as $7x^4$ and $9x^3$, respectively. The sum of the given expressions therefore simplifies to $7x^4+9x^3$.

Choice A is incorrect and may result from adding the coefficients and the exponents in the given expressions. Choice B is incorrect and may result from adding the exponents as well as the coefficients of the like terms in the given expressions. Choice C is incorrect and may result from multiplying, rather than adding, the coefficients of the like terms in the given expressions.

▼ 2.19 — B

Choice B is correct. Expanding all terms yields $(x - 11y)(2x - 3y) - 12y(-2x + 3y)$, which is equivalent to $2x^2 - 22xy - 3xy + 33y^2 + 24xy - 36y^2$. Combining like terms gives $2x^2 - xy - 3y^2$.

Choice A is incorrect and may be the result of using the sums of the coefficients of the existing x and y terms as the coefficients of the x and y terms in the new expressions. Choice C is incorrect and may be the result of incorrectly combining like terms. Choice D is incorrect and may be the result of using the incorrect sign in front of the 12y term.

▼ 2.20 — C

Choice C is correct. The first expression $(1.5x-2.4)^2$ can be rewritten as $(1.5x-2.4)(1.5x-2.4)$. Applying the distributive property to this product yields $(2.25x^2-3.6x-3.6x+5.76)-(5.2x^2-6.4)$. This difference can be rewritten as $(2.25x^2-3.6x-3.6x+5.76)+(-1)(5.2x^2-6.4)$. Distributing the factor of -1 through the second expression yields $2.25x^2-3.6x-3.6x+5.76-5.2x^2+6.4$. Regrouping like terms, the expression becomes $(2.25x^2-5.2x^2)+(-3.6x-3.6x)+(5.76+6.4)$. Combining like terms yields $-2.95x^2-7.2x+12.16$.

Choices A, B, and D are incorrect and likely result from errors made when applying the distributive property or combining the resulting like terms.

▼ 2.21 — B

Choice B is correct. The first and last terms of the polynomial are both squares such that $4a^2=(2a)^2$ and $25b^2=(5b)^2$. The second term is twice the product of the square root of the first and last terms: $20ab=2(2a)(5b)$. Therefore, the polynomial is the square of a binomial such that $4a^2+20ab+25b^2=(2a+5b)^2$, and $(2a+5b)$ is a factor.

Choice A is incorrect and may be the result of incorrectly factoring the polynomial. Choice C is incorrect and may be the result of dividing the second and third terms of the polynomial by their greatest common factor. Choice D is incorrect and may be the result of not factoring the coefficients.

▼ 2.22 — ?

The correct answer is 32. The sum of the given expressions is $(-2x^2+x+31)+(3x^2+7x-8)$. Combining like terms yields $x^2+8x+23$. Based on the form of the given equation, $a=1$, $b=8$, and $c=23$. Therefore, $a+b+c=32$.

Alternate approach: Because $a+b+c$ is the value of ax^2+bx+c when $x=1$, it is possible to first make that substitution into each polynomial before adding them. When $x=1$, the

first polynomial is equal to $-2+1+31=30$ and the second polynomial is equal to $3+7-8=2$. The sum of 30 and 2 is 32.

▼ 2.23 — B

Choice B is correct. In the given expression, $2(x-\frac{9}{2})$ can be rewritten as $2x-2(\frac{9}{2})$, or $2x-9$. Applying the distributive property to the expression $(2x-9)(x+8)$ yields $2x(x+8)-9(x+8)$, or $2x^2+16x-9x-72$. Combining like terms in this expression yields $2x^2+7x-72$. Thus, $2(x-\frac{9}{2})(x+8)$ is equivalent to $2x^2+7x-72$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.24 — D

Choice D is correct. It's given that $gx=\frac{3}{5}x+\frac{7}{6}$ and $hx=6x-5$. Substituting $\frac{3}{5}x+\frac{7}{6}$ for gx and $6x-5$ for hx in the expression $gx\cdot hx$ yields $\frac{3}{5}x+\frac{7}{6}6x-5$. This expression can be rewritten as $\frac{3}{5}x6x-5+\frac{7}{6}6x-5$, or $\frac{18x^2}{5}-3x+7x-\frac{35}{6}$, which is equivalent to $\frac{18x^2}{5}+4x-\frac{35}{6}$.

Choice A is incorrect. This expression is equivalent to $\frac{3}{5}x6x+\frac{7}{6}-5$, not $\frac{3}{5}x+\frac{7}{6}6x-5$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This expression is equivalent to $\frac{3}{5}x-\frac{7}{6}6x+5$, not $\frac{3}{5}x+\frac{7}{6}6x-5$.

▼ 2.25 — D

Choice D is correct. For $x>1$ and $y>1$, $x^{\frac{1}{3}}$ and $y^{\frac{1}{2}}$ are equivalent to $\sqrt[3]{x}$ and $\sqrt{y}$, respectively. Also, x^{-2} and y^{-1} are equivalent to $\frac{1}{x^2}$ and $\frac{1}{y}$, respectively. Therefore, the given expression can be rewritten as $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$.

Choices A, B, and C are incorrect because these choices are not equivalent to the given expression for $x>1$ and $y>1$.

For example, for $x=2$ and $y=2$, the value of the given expression is $2^{-\frac{5}{6}}$; the values of the choices, however, are $2^{-\frac{1}{3}}$, $2^{\frac{5}{6}}$, and 1, respectively.

▼ 2.26 — C

Choice C is correct. Since the expression x^2+6x-7 can be factored as $(x+7)(x-1)$, the given equation can be rewritten as $\frac{(x+7)(x-1)}{x+7}=ax+d$. Since $x\neq -7$, $x+7$ is also not equal to 0, so both the numerator and denominator of $\frac{(x+7)(x-1)}{x+7}$ can be divided by $x+7$. This gives $x-1=ax+d$. Equating the coefficient of x on each side of the equation gives $a=1$. Equating the constant terms gives $d=-1$. The sum is $1+(-1)=0$.

Choice A is incorrect and may result from incorrectly simplifying the equation. Choices B and D are incorrect. They are the values of d and a, respectively, not $a+d$.

▼ 2.27 — D

Choice D is correct. Since $x^3-9x=x(x+3)(x-3)$ and $x^2-2x-3=(x+1)(x-3)$, the fraction $\frac{f(x)}{g(x)}$ can be written as $\frac{x(x+3)(x-3)}{(x+1)(x-3)}$. It is given that $x>3$, so the common factor $x-3$ is not equal to 0. Therefore, the fraction can be further simplified to $\frac{x(x+3)}{x+1}$.

Choice A is incorrect. The expression $\frac{1}{x+1}$ is not equivalent to $\frac{f(x)}{g(x)}$ because at $x=0$, $\frac{1}{x+1}$ as a value of 1 and $\frac{f(x)}{g(x)}$ has a value of 0.

Choice B is incorrect and results from omitting the factor x in the factorization of $f(x)$.

Choice C is incorrect and may result from incorrectly factoring $g(x)$ as $(x+1)(x+3)$ instead of

$(x+1)(x-3)$.

▼ 2.28 — 27

The correct answer is 27. The given expression $\frac{p}{x} + \frac{w}{x+51}$ can be rewritten with the common denominator $x(x+51)$. To add the two terms of this expression, $\frac{p}{x}$ can be rewritten as $\frac{p(x+51)}{x(x+51)}$ and $\frac{w}{(x+51)}$ can be rewritten as $\frac{wx}{x(x+51)}$. Therefore, the expression $\frac{p}{x} + \frac{w}{(x+51)}$ can be rewritten as $\frac{p(x+51)}{x(x+51)} + \frac{wx}{x(x+51)}$, which is equivalent to $\frac{px+51p+wx}{x(x+51)}$, or $\frac{(p+w)x+51p}{x(x+51)}$. Since $\frac{29x+102}{x(x+51)}$ is equivalent to $\frac{(p+w)x+51p}{x(x+51)}$ and these expressions have equivalent denominators, these expressions must also have equivalent numerators. It follows that $29x + 102 = (p+w)x + 51p$. In this equation, the x-coefficients must be equal and the constant terms must be equal. Therefore, $p + w = 29$ and $51p = 102$. Dividing both sides of $51p = 102$ by 51 yields $p = 2$. Substituting 2 for p in $p + w = 29$ yields $2 + w = 29$. Subtracting 2 from both sides of this equation yields $w = 27$. Therefore, the value of w is 27.

▼ 2.29 — ?

The correct answer is $\frac{7}{6}$. The value of $\frac{a}{b}$ can be found by first rewriting the left-hand side of the given equation as $\frac{x^{\frac{5}{2}}}{x^{\frac{4}{3}}}$. Using the properties of exponents, this expression can be rewritten as $x^{\left(\frac{5}{2}-\frac{4}{3}\right)}$. This expression can be rewritten by subtracting the fractions in the exponent, which yields $x^{\frac{7}{6}}$. Thus, $\frac{a}{b}$ is $\frac{7}{6}$. Note that 7/6, 1.166, and 1.167 are examples of ways to enter a correct answer.

▼ 2.30 — 8

The correct answer is 8. Since each term of the given expression, $2x^3 + 42x^2 + 208x$, has a factor of $2x$, the expression can be rewritten as $2xx^2 + 2x21x + 2x104$, or $2xx^2 + 21x + 104$. Since the values 8 and 13 have a sum of 21 and a product of 104, the expression $x^2 + 21x + 104$ can be factored as $x + 8x + 13$. Therefore, the given expression can be factored as $2xx + 8x + 13$. It follows that the factors of the given expression are 2, x, $x + 8$, and $x + 13$. Of these factors, only $x + 8$ and $x + 13$ are of the form $x + b$, where b is a positive

constant. Therefore, the possible values of b are 8 and 13. Thus, the smallest possible value of b is 8.

▼ 2.31 — D

Choice D is correct. The numerator of the given expression can be rewritten in terms of the denominator, $x-3$, as follows: $x^2-2x-5=x^2-3x+x-3-2$, which is equivalent to $x(x-3)+(x-3)-2$. So the given expression is equivalent to $\frac{x(x-3)+(x-3)-2}{x-3}=\frac{x(x-3)}{x-3}+\frac{x-3}{x-3}-\frac{2}{x-3}$. Since the given expression is defined for $x\neq 3$, the expression can be rewritten as $x+1-\frac{2}{x-3}$.

Long division can also be used as an alternate approach. Choices A, B, and C are incorrect and may result from errors made when dividing the two polynomials or making use of structure.

▼ 2.32 — .0177, .0178, 4/225

The correct answer is $\frac{4}{225}$. An expression of the form $\sqrt[k]{a}$, where k is an integer greater than 1 and $a \geq 0$, is equivalent to $a^{\frac{1}{k}}$. Therefore, the given expression, where $n > 1$, is equivalent to $70n^{\frac{1}{5}}\,70n^{\frac{1}{6}}{}^{2}$. Applying properties of exponents, this expression can be rewritten as $70n^{\frac{1}{5}}\,70n^{\frac{1}{6}\cdot 2}$, or $70n^{\frac{1}{5}}\,70n^{\frac{1}{3}}$, which can be rewritten as $70n^{\frac{1}{5}+\frac{1}{3}}$, or $70n^{\frac{8}{15}}$. It's given that the expression $\sqrt[5]{70n}\,\sqrt[6]{70n}^{2}$ is equivalent to $70n^{30x}$, where $n > 1$. It follows that $70n^{\frac{8}{15}}$ is equivalent to $70n^{30x}$. Therefore, $\frac{8}{15} = 30x$. Dividing both sides of this equation by 30 yields $\frac{8}{450} = x$, or $\frac{4}{225} = x$. Thus, the value of x for which the given expression is equivalent to $70n^{30x}$, where $n > 1$, is $\frac{4}{225}$. Note that 4/225, .0177, .0178, 0.017, and 0.018 are examples of ways to enter a correct answer.

▼ 2.33 — C

Choice C is correct. If the equation is true for all x, then the expressions on both sides of the equation will be equivalent. Multiplying the polynomials on the left-hand side of the equation gives $5ax^3-abx^2+4ax+15x^2-3bx+12$. On the right-hand side of the equation, the only x^2-term is $-9x^2$. Since the expressions on both sides of the equation are equivalent, it

follows that $-abx^2+15x^2=-9x^2$, which can be rewritten as $(-ab+15)x^2=-9x^2$. Therefore, $-ab+15=-9$, which gives $ab=24$.

Choice A is incorrect. If $ab=18$, then the coefficient of x^2 on the left-hand side of the equation would be $-18+15=-3$, which doesn't equal the coefficient of x^2, -9, on the right-hand side. Choice B is incorrect. If $ab=20$, then the coefficient of x^2 on the left-hand side of the equation would be $-20+15=-5$, which doesn't equal the coefficient of x^2, -9, on the right-hand side. Choice D is incorrect. If $ab=40$, then the coefficient of x^2 on the left-hand side of the equation would be $-40+15=-25$, which doesn't equal the coefficient of x^2, -9, on the right-hand side.

▼ 2.34 — ?

The correct answer is 6632. Applying the distributive property to the expression yields $(7532+100y^2)+(100y^2-1100)$. Then adding together $7532+100y^2$ and $100y^2-1100$ and collecting like terms results in $200y^2+6432$. This is written in the form ay^2+b, where $a=200$ and $b=6432$. Therefore $a+b=200+6432=6632$.

▼ 2.35 — D

Choice D is correct. Two fractions can be added together when they have a common denominator. Since $k > 0$, multiplying the second term in the given expression by $\frac{k}{k}$ yields $\frac{42akk}{k}$, which is equivalent to $\frac{42ak^2}{k}$. Therefore, the expression $\frac{42a}{k}+42ak$ can be written as $\frac{42a}{k}+\frac{42ak^2}{k}$ which is equivalent to $\frac{42a+42ak^2}{k}$. Since each term in the numerator of this expression has a factor of $42a$, the expression $\frac{42a+42ak^2}{k}$ can be rewritten as $\frac{42a1+42ak^2}{k}$, or $\frac{42a1+k^2}{k}$, which is equivalent to $\frac{42a(k^2+1)}{k}$.

Choice A is incorrect. This expression is equivalent to $\frac{42a}{k}+\frac{42a}{k}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This expression is equivalent to $\frac{42a}{k}+42a$.

▼ 2.36 — C

Choice C is correct. For positive values of a, $\sqrt[N]{a} = a^{\frac{1}{N}}$. Therefore, the given expression, $\frac{\sqrt[3]{n^{17}p^5}}{5n^2\sqrt[8]{p^8}}$, can be rewritten as $\frac{n^{17}p^{5\frac{1}{3}}}{5n^2p^{8\frac{1}{8}}}$. Applying properties of exponents, this can be rewritten as $\frac{n^{\frac{17}{3}}p^{\frac{5}{3}}}{5n^2p^{\frac{8}{8}}}$, or $\frac{n^{\frac{17}{3}}p^{\frac{5}{3}}}{5n^2p}$. Dividing the numerator and the denominator of this expression by n yields $\frac{n^{\frac{14}{3}}p^{\frac{5}{3}}}{5np}$, which is statement I. Dividing the numerator and the denominator of this expression by np yields $\frac{n^{\frac{11}{3}}p^{\frac{2}{3}}}{5}$. This expression is equivalent to $\frac{n^{11}p^{2\frac{1}{3}}}{5}$. It follows that this expression is equivalent to $\frac{\sqrt[3]{n^{11}p^2}}{5}$, which is statement II. Therefore, for all positive values of n and p, the given expression is equivalent to I and II.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.37 — .09, 9/100

The correct answer is $.09$. It's given that the expression $0.36x^2 + 0.63x + 1.17$ can be rewritten as $a4x^2 + 7x + 13$. Applying the distributive property to the expression $a4x^2 + 7x + 13$ yields $4ax^2 + 7ax + 13a$. Therefore, $0.36x^2 + 0.63x + 1.17$ can be rewritten as $4ax^2 + 7ax + 13a$. It follows that in the expressions $0.36x^2 + 0.63x + 1.17$ and $4ax^2 + 7ax + 13a$, the coefficients of x^2 are equivalent, the coefficients of x are equivalent, and the constant terms are equivalent. Therefore, $0.36 = 4a$, $0.63 = 7a$, and $1.17 = 13a$. Solving any of these equations for a yields the value of a. Dividing both sides of the equation $0.36 = 4a$ by 4 yields $0.09 = a$. Therefore, the value of a is 0.09. Note that .09 and 9/100 are examples of ways to enter a correct answer.

▼ 2.38 — D

Choice D is correct. If $k - x$ is a factor of the expression $-x^2 + \frac{1}{29}nk^2$, then the expression can be written as $k - xax + b$, where a and b are constants. This expression can be rewritten as $akx + bk - ax^2 - bx$, or $-ax^2 + ak - bx + bk$. Since this expression is equivalent to $-x^2 + \frac{1}{29}nk^2$, it follows that $-a = -1$, $ak - b = 0$, and $bk = \frac{1}{29}nk^2$. Dividing each side of the equation $-a = -1$ by -1 yields $a = 1$. Substituting 1 for a in the equation $ak - b = 0$ yields $k - b = 0$. Adding b to each side of this equation yields $k = b$. Substituting k for b in the equation $bk = \frac{1}{29}nk^2$ yields $k^2 = \frac{1}{29}nk^2$. Since k is positive, dividing each side of this equation by k^2 yields $1 = \frac{1}{29}n$. Multiplying each side of this equation by 29 yields $29 = n$.

Alternate approach: The expression $x^2 - y^2$ can be written as $x - yx + y$, which is a difference of two squares. It follows that $\frac{1}{29}nk^2 - x^2$ is equivalent to $\sqrt{\frac{1}{29}n}k - x\sqrt{\frac{1}{29}n}k + x$. It's given that $k - x$ is a factor of $-x^2 + \frac{1}{29}nk^2$, so the factor $\sqrt{\frac{1}{29}n}k - x$ is equal to $k - x$. Adding x to both sides of the equation $\sqrt{\frac{1}{29}n}k - x = k - x$ yields $\sqrt{\frac{1}{29}n}k = k$. Since k is positive, dividing both sides of this equation by k yields $\sqrt{\frac{1}{29}n} = 1$. Squaring both sides of this equation yields $\frac{1}{29}n = 1$. Multiplying both sides of this equation by 29 yields $n = 29$.

Choice A is incorrect. This value of n gives the expression $-x^2 + \frac{1}{29}\text{-}29k^2$, or $-x^2 - k^2$. This expression doesn't have $k - x$ as a factor.

Choice B is incorrect. This value of n gives the expression $-x^2 + \frac{1}{29}\text{-}\frac{1}{29}k^2$, or $-x^2 + \text{-}\frac{1}{841}k^2$. This expression doesn't have $k - x$ as a factor.

Choice C is incorrect. This value of n gives the expression $-x^2 + \frac{1}{29}\frac{1}{29}k^2$, or $-x^2 + \frac{1}{841}k^2$. This expression doesn't have $k - x$ as a factor.

▼ 2.39 — C

Choice C is correct. Factoring the denominator in the second term of the given expression gives $\frac{y + 12}{x - 8} + \frac{yx - 8}{xyx - 8}$. This expression can be rewritten with common denominators by multiplying the first term by $\frac{xy}{xy}$, giving $\frac{xyy + 12}{xyx - 8} + \frac{yx - 8}{xyx - 8}$. Adding these two terms yields $\frac{xyy + 12 + yx - 8}{xyx - 8}$. Using the distributive property to rewrite this expression gives $\frac{xy^2 + 12xy + xy - 8y}{x^2y - 8xy}$. Combining the like terms in the numerator of this expression gives $\frac{xy^2 + 13xy - 8y}{x^2y - 8xy}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.40 — C

Choice C is correct. To express the sum of the two rational expressions on the left-hand side of the equation as the single rational expression on the right-hand side of the equation, the expressions on the left-hand side must have the same denominator. Multiplying the first expression by $\frac{x+5}{x-5}$ results in $\frac{2(x+5)}{(x-2)(x+5)}$, and multiplying the second expression by $\frac{x-2}{x-2}$ results in $\frac{3(x-2)}{(x-2)(x+5)}$, so the given equation can be rewritten as

$\frac{2(x+5)}{(x-2)(x+5)}+\frac{3(x-2)}{(x-2)(x+5)}=\frac{rx+t}{(x-2)(x+5)}$, or $\frac{2x+10}{(x-2)(x+5)}+\frac{3x-6}{(x-2)(x+5)}=\frac{rx+t}{(x-2)(x+5)}$. Since the two rational expressions on the left-hand side of the equation have the same denominator as the rational expression on the right-hand side of the equation, it follows that $(2x+10)+(3x-6)=rx+t$. Combining like terms on the left-hand side yields $5x+4=rx+t$, so it follows that $r=5$ and $t=4$. Therefore, the value of rt is $(5)(4)=20$.

Choice A is incorrect and may result from an error when determining the sign of either r or t. Choice B is incorrect and may result from not distributing the 2 and 3 to their respective terms in $\frac{2(x+5)}{(x-2)(x+5)}+\frac{3(x-2)}{(x-2)(x+5)}=\frac{rx+t}{(x-2)(x+5)}$. Choice D is incorrect and may result from a calculation error.

▼ 2.41 — 15.5, 31/2

The correct answer is 15.5. It's given that $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers. The expression $(3x^2 + a)(2x^2 + b)$ can be rewritten as $6x^4 + 3bx^2 + 2ax^2 + ab$, or $6x^4 + (3b + 2a)x^2 + ab$. Therefore, the expression $6x^4 + (3b + 2a)x^2 + ab$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3b + 2a = 31$ and $ab = 35$. Since a and b are positive integers and $ab = 35$, the possible values of a and b are 1 and 35, 5 and 7, 7 and 5, and 35 and 1, respectively. Of these options, the only pair that satisfies the equation $3b + 2a = 31$ is $a = 5$ and $b = 7$, since $3(7) + 2(5) = 31$. It's also given that $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. The expression $(3x^2 + c)(2x^2 + d)$ can be rewritten as $6x^4 + 3dx^2 + 2cx^2 + cd$, or $6x^4 + (3d + 2c)x^2 + cd$. Therefore, the expression $6x^4 + (3d + 2c)x^2 + cd$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3d + 2c = 31$ and $cd = 35$. Since it's given that c and d are positive nonintegers, it follows that c and d are not zero. Thus, dividing both sides of the equation $cd = 35$ by c yields $d = \frac{35}{c}$. Substituting $\frac{35}{c}$ for d in the equation $3d + 2c = 31$ yields $3\frac{35}{c} + 2c = 31$, or $\frac{105}{c} + 2c = 31$. Multiplying both sides of this equation by c yields $c\frac{105}{c} + 2c = c(31)$, or $105 + 2c^2 = 31c$. Subtracting $31c$ from both sides of this equation yields $105 + 2c^2 - 31c = 0$, or $2c^2 - 31c + 105 = 0$. This equation can be rewritten as $2c^2 - 10c - 21c + 105 = 0$. Since the first two terms of the expression $2c^2 - 10c - 21c + 105$ have a common factor of $2c$ and the last two terms of the expression $2c^2 - 10c - 21c + 105$ have a common factor of 21, the expression $2c^2 - 10c - 21c + 105$ can be

rewritten as $2c(c - 5) - 21(c - 5)$. Since each term of this expression has a common factor of $(c - 5)$, the expression $2c(c - 5) - 21(c - 5)$ can be rewritten as $(c - 5)(2c - 21)$. Thus, the equation $2c^2 - 31c + 105 = 0$ can be rewritten as $(c - 5)(2c - 21) = 0$. It follows that $c = 5$ or $c = \frac{21}{2}$. Since c is a positive noninteger, it follows that $c = \frac{21}{2}$, or $c = 10.5$. Substituting 5 for a and 10.5 for c into the expression $a + c$ yields $5 + 10.5$, or 15.5. Therefore, the value of $a + c$ is 15.5. Note that 15.5 and 31/2 are examples of ways to enter a correct answer.

Alternate approach: The correct answer is 15.5. It's given that $6x^4 + 31x^2 + 35$ can be rewritten as $3x^2 + a2x^2 + b$, where a and b are positive integers. The expression $3x^2 + a2x^2 + b$ can be rewritten as $6x^4 + 3bx^2 + 2ax^2 + ab$, or $6x^4 + (3b + 2a)x^2 + ab$. Therefore, the expression $6x^4 + 3b + 2ax^2 + ab$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3b + 2a = 31$ and $ab = 35$. Since it's given that a and b are positive integers, it follows that a and b are not zero. Thus, dividing both sides of the equation $ab = 35$ by a yields $b = \frac{35}{a}$. Substituting $\frac{35}{a}$ for b in the equation $3b + 2a = 31$ yields $3\frac{35}{a} + 2a = 31$, or $\frac{105}{a} + 2a = 31$. Multiplying both sides of this equation by a yields $105 + 2a^2 = 31a$. Subtracting $31a$ from both sides of this equation yields $2a^2 - 31a + 105 = 0$. This equation can be rewritten as $2a^2 - 10a - 21a + 105 = 0$. Since the first two terms of the expression $2a^2 - 10a - 21a + 105$ have a common factor of $2a$ and the last two terms of the expression $2a^2 - 10a - 21a + 105$ have a common factor of 21, the expression $2a^2 - 10a - 21a + 105$ can be rewritten as $2a(a - 5) - 21(a - 5)$. Since each term of this expression has a common factor of $a - 5$, the expression $2a(a - 5) - 21(a - 5)$ can be rewritten as $(a - 5)(2a - 21)$. Thus, the equation $2a^2 - 31a + 105 = 0$ can be rewritten as $a - 52a - 21 = 0$. It follows that $a = 5$ or $a = \frac{21}{2}$. Since a is a positive integer, it follows that $a = 5$. It's also given that $6x^4 + 31x^2 + 35$ can be rewritten as $3x^2 + c2x^2 + d$, where c and d are positive nonintegers. The expression $3x^2 + c2x^2 + d$ can be rewritten as $6x^4 + 3dx^2 + 2cx^2 + cd$, or $6x^4 + (3d + 2c)x^2 + cd$. Therefore, the expression $6x^4 + (3d + 2c)x^2 + cd$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3d + 2c = 31$ and $cd = 35$. Since it's given that c and d are positive nonintegers, it follows that c and d are not zero. Thus, dividing both sides of the equation $cd = 35$ by c yields $d = \frac{35}{c}$. Substituting $\frac{35}{c}$ for d in the equation $3d + 2c = 31$ yields $3\frac{35}{c} + 2c = 31$, or $\frac{105}{c} + 2c = 31$. Multiplying both sides of this equation by c yields $105 + 2c^2 = 31c$. Subtracting $31c$ from both sides of this equation yields $2c^2 - 31c + 105 = 0$. Since the equations $2a^2 - 31a + 105 = 0$ and $2c^2 - 31c + 105 = 0$ represent the same equation, it follows that a is equal to c. Therefore, $c = 5$ or $c = \frac{21}{2}$. Since c is a positive noninteger, it follows that $c = \frac{21}{2}$, or $c = 10.5$. Substituting 5 for a and 10.5 for c in the expression $a + c$ yields $5 + 10.5$, or 15.5. Therefore, the value of $a + c$ is 15.5. Note that 15.5 and 31/2 are examples of ways to enter a correct answer.

▼ 2.42 — D

Choice D is correct. The expression $\frac{4}{4x-5} - \frac{1}{x+1}$ can be rewritten as $\frac{4}{4x-5} + \frac{-1}{x+1}$. To add the two terms of this expression, the terms can be rewritten with a common denominator. Since $\frac{x+1}{x+1} = 1$, the expression $\frac{4}{4x-5}$ can be rewritten as $\frac{x+14}{x+14x-5}$. Since $\frac{4x-5}{4x-5} = 1$, the expression $\frac{-1}{x+1}$ can be rewritten as $\frac{4x-5-1}{4x-5x+1}$. Therefore, the expression $\frac{4}{4x-5} + \frac{-1}{x+1}$ can be rewritten as $\frac{x+14}{x+14x-5} + \frac{4x-5-1}{4x-5x+1}$, which is equivalent to $\frac{x+14+4x-5-1}{x+14x-5}$. Applying the distributive property to each term of the numerator yields $\frac{4x+4+-4x+5}{x+14x-5}$, or $\frac{4x+-4x+4+5}{x+14x-5}$. Adding like terms in the numerator yields $\frac{9}{x+14x-5}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.43 — .5061, .5062, 41/81

The correct answer is $\frac{41}{81}$. An expression of the form $\sqrt[n]{a^m}$, where m and n are integers greater than 1 and $a \geq 0$, is equivalent to $a^{\frac{m}{n}}$. Therefore, the expression on the left-hand side of the given equation is equivalent to $p^{\frac{2}{3}}$; thus, $p^{\frac{2}{3}} = t^{\frac{9}{7}}$. If $t = p^{3n-1}$, where n is a constant, then substituting p^{3n-1} for t in the equation yields $p^{\frac{2}{3}} = p^{\frac{9}{7}(3n-1)}$. It's given that $p > 1$; therefore, $\frac{2}{3} = \frac{9}{7}(3n-1)$. Multiplying each side of this equation by 21 yields $14 = 27(3n-1)$. Distributing multiplication over subtraction yields $14 = 81n - 27$. Adding 27 to each side of this equation yields $41 = 81n$. Dividing each side of this equation by 81 yields $\frac{41}{81} = n$. Therefore, if $t = p^{3n-1}$, where n is a constant, the value of n is $\frac{41}{81}$. Note that 41/81, .5061, .5062, and 0.506 are examples of ways to enter a correct answer.

▼ 2.44 — D

Choice D is correct. Factoring out the coefficient $\frac{1}{3}$, the given expression can be rewritten as $\frac{1}{3}(x^2-6)$. The expression x^2-6 can be approached as a difference of squares and rewritten as $(x-\sqrt{6})(x+\sqrt{6})$. Therefore, k must be $\sqrt{6}$.

Choice A is incorrect. If k were 2, then the expression given would be rewritten as $\frac{1}{3}(x-2)(x+2)$, which is equivalent to $\frac{1}{3}x^2-\frac{4}{3}$, not $\frac{1}{3}x^2-2$.

Choice B is incorrect. This may result from incorrectly factoring the expression and finding $(x-6)(x+6)$ as the factored form of the expression. Choice C is incorrect. This may result

from incorrectly distributing the $\frac{1}{3}$ and rewriting the expression as $\frac{1}{3}(x^2-2)$.

▼ 2.45 — D

Choice D is correct. The expression $\left(a+\frac{b}{2}\right)^2$ can be rewritten as $\left(a+\frac{b}{2}\right)\left(a+\frac{b}{2}\right)$. Using the distributive property, the expression yields $\left(a+\frac{b}{2}\right)\left(a+\frac{b}{2}\right)=a^2+\frac{ab}{2}+\frac{ab}{2}+\frac{b^2}{4}$. Combining like terms gives $a^2+ab+\frac{b^2}{4}$.

Choices A, B, and C are incorrect and may result from errors using the distributive property on the given expression or combining like terms.

▼ 2.46 — 361/8, 45.12, 45.13

The correct answer is $\frac{361}{8}$. The rational exponent property is $\sqrt[n]{y^m}=y^{\frac{m}{n}}$, where $y>0$, m and n are integers, and $n>0$. This property can be applied to rewrite the given expression $6\sqrt[5]{3^5x^{45}}\cdot\sqrt[8]{2^8x}$ as $63^{\frac{5}{5}}x^{\frac{45}{5}}2^{\frac{8}{8}}x^{\frac{1}{8}}$, or $63x^9\,2x^{\frac{1}{8}}$. This expression can be rewritten by multiplying the constants, which gives $36x^9x^{\frac{1}{8}}$. The multiplication exponent property is $y^n\cdot y^m=y^{n+m}$, where $y>0$. This property can be applied to rewrite the expression $36x^9x^{\frac{1}{8}}$ as $36x^{9+\frac{1}{8}}$, or $36x^{\frac{73}{8}}$. Therefore, $6\sqrt[5]{3^5x^{45}}\cdot\sqrt[8]{2^8x}=36x^{\frac{73}{8}}$. It's given that $6\sqrt[5]{3^5x^{45}}\cdot\sqrt[8]{2^8x}$ is equivalent to ax^b; therefore, $a=36$ and $b=\frac{73}{8}$. It follows that $a+b=36+\frac{73}{8}$. Finding a common denominator on the right-hand side of this equation gives $a+b=\frac{288}{8}+\frac{73}{8}$, or $a+b=\frac{361}{8}$. Note that 361/8, 45.12, and 45.13 are examples of ways to enter a correct answer.

▼ 2.47 — B

Choice B is correct. The given expression can be factored by first finding two values whose sum is 20 and whose product is $3\text{-}63$, or -189. Those two values are 27 and -7. It follows that the given expression can be rewritten as $3x^2+27x-7x-63$. Since the first two terms of this expression have a common factor of $3x$ and the last two terms of this expression have a common factor of -7, this expression can be rewritten as $3x(x+9)-7(x+9)$. Since the two terms of this expression have a common factor of $x+9$, it can be rewritten as $(3x-7)(x+9)$.

Therefore, expression II, $3x - 7$, is a factor of $3x^2 + 20x - 63$, but expression I, $x - 9$, is not a factor of $3x^2 + 20x - 63$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.48 — .2916, .2917, 7/24

The correct answer is $\frac{7}{24}$. An expression of the form $\sqrt[n]{a^m}$, where *m* and *n* are integers greater than 1 and $a \geq 0$, is equivalent to $a^{\frac{m}{n}}$. Therefore, the expression on the right-hand side of the given equation, $\sqrt[3]{4^7}$, is equivalent to $4^{\frac{7}{3}}$. Thus, $4^{8c} = 4^{\frac{7}{3}}$. It follows that $8c = \frac{7}{3}$. Dividing both sides of this equation by 8 yields $c = \frac{7}{24}$. Note that 7/24, .2916, .2917, 0.219, and 0.292 are examples of ways to enter a correct answer.

▼ 2.49 — A

Choice A is correct. The given expression is equivalent to $9x^4 + 8x^3 - 7x^3 - 5x^2 - 3x^2 - x$. By combining like terms, this can be written as $9x^4 + x^3 - 8x^2 - x$. This expression is in the form $9x^4 + x^3 + ax^2 - x$, where *a* is a constant. Therefore, the value of *a* is -8.

Choice B is incorrect. This may result from adding $-5x^2$ and $3x^2$ rather than $-5x^2$ and $-3x^2$.

Choice C is incorrect. This may result from adding $5x^2$ and $-3x^2$ rather than $-5x^2$ and $-3x^2$.

Choice D is incorrect. This may result from adding $5x^2$ and $3x^2$ rather than $-5x^2$ and $-3x^2$.

▼ 2.50 — B

Choice B is correct. Using the distributive property, the first given expression can be rewritten as $6x^2 + 3px + 24 – 16px – 64x + 24$, and then rewritten as $6x^2 + (3p – 16p – 64)x + 24$. Since the expression $6x^2 + (3p – 16p – 64)x + 24$ is equivalent to $6x^2 – 155x + 24$, the coefficients of the x terms from each expression are equivalent to each other; thus $3p – 16p – 64 = –155$. Combining like terms gives $–13p – 64 = –155$. Adding 64 to both sides of the equation gives $–13p = –71$. Dividing both sides of the equation by –13 yields $p = 7$.

Choice A is incorrect. If $p = –3$, then the first expression would be equivalent to $6x^2 – 25x + 24$. Choice C is incorrect. If $p = 13$, then the first expression would be equivalent to $6x^2 –$

233x + 24. Choice D is incorrect. If p = 155, then the first expression would be equivalent to $6x^2 - 2{,}079x + 24$.

▼ 2.51 — A

Choice A is correct. If the given expression is equivalent to $x+b$, then $\frac{x^2-c}{x-b}=x+b$, where x isn't equal to b. Multiplying both sides of this equation by $x-b$ yields $x^2-c=(x+b)(x-b)$. Since the right-hand side of this equation is in factored form for the difference of squares, the value of c must be a perfect square. Only choice A gives a perfect square for the value of c.

Choices B, C, and D are incorrect. None of these values of c produces a difference of squares. For example, when 6 is substituted for c in the given expression, the result is $\frac{x^2-6}{x-b}$. The expression x^2-6 can't be factored with integer values, and therefore $\frac{x^2-6}{x-b}$ isn't equivalent to $x+b$.

▼ 2.52 — D

Choice D is correct. The product of $x^{-6}y^3z^5$ and $x^4z^5+y^8z^{-7}$ can be represented by the expression $x^{-6}y^3z^5x^4z^5+y^8z^{-7}$. Applying the distributive property to this expression yields $x^{-6}y^3z^5x^4z^5+x^{-6}y^3z^5y^8z^{-7}$, or $x^{-6}x^4y^3z^5z^5+x^{-6}y^3y^8z^5z^{-7}$. This expression is equivalent to $x^{-6+4}y^3z^{5+5}+x^{-6}y^{3+8}z^{5-7}$, or $x^{-2}y^3z^{10}+x^{-6}y^{11}z^{-2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.53 — D

Choice D is correct. Since each choice has a term of $3x^2$, which can be written as $3xx$, and each choice has a term of $14b$, which can be written as $7 \cdot 2b$, the expression that has a factor of $x + 2b$, where b is a positive integer constant, can be represented as $3x + 7x + 2b$. Using the distributive property of multiplication, this expression is equivalent to $3xx + 2b + 7x + 2b$, or $3x^2 + 6xb + 7x + 14b$. Combining the x-terms in this expression yields $3x^2 + 7 + 6bx + 14b$.

It follows that the coefficient of the x-term is equal to $7 + 6b$. Thus, from the given choices, $7 + 6b$ must be equal to 7, 28, 42, or 49. Therefore, $6b$ must be equal to 0, 21, 35, or 42, respectively, and b must be equal to $\frac{0}{6}$, $\frac{21}{6}$, $\frac{35}{6}$, or $\frac{42}{6}$, respectively. Of these four values of b, only $\frac{42}{6}$, or 7, is a positive integer. It follows that $7 + 6b$ must be equal to 49 because this is the only choice for which the value of b is a positive integer constant. Therefore, the expression that has a factor of $x + 2b$ is $3x^2 + 49x + 14b$.

Choice A is incorrect. If this expression has a factor of $x + 2b$, then the value of b is 0, which isn't positive.

Choice B is incorrect. If this expression has a factor of $x + 2b$, then the value of b is $\frac{21}{6}$, which isn't an integer.

Choice C is incorrect. If this expression has a factor of $x + 2b$, then the value of b is $\frac{35}{6}$, which isn't an integer.

▼ 2.54 — C

Choice C is correct. Using the property that $\sqrt{x}\sqrt{y} = \sqrt{xy}$ for positive numbers x and y, with $x = (a + c)^3$ and $y = a + c$, it follows that $\sqrt{(a+c)^3} \cdot \sqrt{a+c} = \sqrt{(a+c)^4}$. By rewriting $(a + c)^4$ as $((a + c)^2)^2$, it is possible to simplify the square root expression as follows: $\sqrt{((a+c)^2)^2} = (a+c)^2 = a^2+2ac+c^2$.

Choice A is incorrect and may be the result of $\sqrt{(a+c)^3} \div \sqrt{(a+c)}$. Choice B is incorrect and may be the result of incorrectly rewriting $(a + c)^2$ as $a^2 + c^2$. Choice D is incorrect and may be the result of incorrectly applying properties of exponents.

▼ 2.55 — -419

The correct answer is -419. It's given that the expression $3x - 2319x + 6$ is equivalent to the expression $ax^2 + bx + c$, where a, b, and c are constants. Applying the distributive property to the given expression, $3x - 2319x + 6$, yields $3x19x + 3x6 - 2319x - 236$, which can be rewritten as $57x^2 + 18x - 437x - 138$. Combining like terms yields $57x^2 - 419x - 138$. Since this expression is equivalent to $ax^2 + bx + c$, it follows that the value of b is -419.

▼ 2.56 — A

Choice A is correct. Expanding each squared binomial in the given expression yields $-14(25x^2 - 20x + 4) + 6(25x^2 - 30x + 9)$. Applying the distributive property to both terms of this expression yields $(-350x^2 + 280x - 56) + (150x^2 - 180x + 54)$. Combining like terms in this expression yields $-200x^2 + 100x - 2$. Since the given expression can be rewritten as $\frac{a}{6}x^2 + \frac{b}{6}x + \frac{c}{6}$, it follows that $\frac{a}{6} = -200$, $\frac{b}{6} = 100$, and $\frac{c}{6} = -2$. Multiplying both sides of each of these equations by 6 yields $a = -1{,}200$, $b = 600$, and $c = -12$, respectively. Substituting $-1{,}200$ for a, 600 for b, and -12 for c in the expression $a + b + c$ yields $-1{,}200 + 600 - 12$, or -612.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $\frac{a}{6} + \frac{b}{6} + \frac{c}{6}$, not $a + b + c$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.57 — C

Choice C is correct. Factoring –1 from the second, third, and fourth terms gives $x^2 - c^2 - 2cd - d^2 = x^2 - (c^2 + 2cd + d^2)$. The expression $c^2 + 2cd + d^2$ is the expanded form of a perfect square: $c^2 + 2cd + d^2 = (c + d)^2$. Therefore, $x^2 - (c^2 + 2cd + d^2) = x^2 - (c + d)^2$. Since $a = c + d$, $x^2 - (c + d)^2 = x^2 - a^2$. Finally, because $x^2 - a^2$ is the difference of squares, it can be expanded as $x^2 - a^2 = (x + a)(x - a)$.

Choices A and B are incorrect and may be the result of making an error in factoring the difference of squares $x^2 - a^2$. Choice D is incorrect and may be the result of incorrectly combining terms.

▼ 2.58 — D

Choice D is correct. It's given that $4x^2 + bx - 45$ can be rewritten as $hx + kx + j$. The expression $hx + kx + j$ can be rewritten as $hx^2 + jhx + kx + kj$, or $hx^2 + jh + kx + kj$. Therefore, $hx^2 + jh + kx + kj$ is equivalent to $4x^2 + bx - 45$. It follows that $kj = -45$. Dividing each side of this equation by k yields $j = \frac{-45}{k}$. Since j is an integer, $-\frac{45}{k}$ must be an integer. Therefore, $\frac{45}{k}$ must also be an integer.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.